Eigenvalues, Eigenvectors, and Spectral Intuition

Chapter 6 | Book 1: Mathematical Foundations | PU-B01-C06

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Controlled chapter candidate v0.1.0 | Review edition

This chapter preserves the source's eleven-part teaching sequence and its six native displayed equations, repairs damaged inline mathematics, and adds worked material aligned with the tested M1_N06 v0.1.0 notebook. The original master and notebook remain unchanged. A controlled candidate is versioned material for review, not public-release authorization.

Prerequisites: matrix multiplication, vector spaces, bases, rank, orthogonality and norms from Lessons 4–5. All datasets and sector coefficients below are synthetic teaching examples, not empirical policy estimates. Real column vectors are used unless a complex example is explicitly introduced.

6.1 Discovering the Dominant Directions of a System

Canonical source: Chapter 6 in the Book 1 complete master. Controlled companion: M1_N06_eigenvalues_eigenvectors_spectral_intuition_v0.1.0.ipynb. Error-free VS Code and Colab execution has been reported by the user; independent mathematical review and chapter approval remain separate gates.

Some nonzero directions are transformed into scalar multiples of themselves. Positive real eigenvalues preserve orientation, negative ones reverse it, and zero maps the vector to zero. These directions reveal the transformation's hidden structure.

Lesson 5 expressed vectors through coordinates in a basis. This lesson asks whether a basis can be chosen so that a transformation acts independently on its coordinates. A basis of eigenvectors does exactly that. Not every matrix has such a basis; understanding the limitation is part of the method.

6.2 Learning Objectives

The reader will define and verify eigenpairs, interpret eigenvalues geometrically, understand diagonalization and matrix powers, estimate dominant eigenpairs, analyze spectral stability, and connect spectral methods to covariance, networks and dynamic decision systems.

Explain each central example in three ways: geometrically, algebraically and computationally. Reject zero eigenvectors, distinguish residual accuracy from dominance, and separate mathematical structure from a decision recommendation.

6.3 Opening Story — A Shock Moving Through the Economy

A fuel-price shock can affect transport, agriculture, food prices and household welfare. As an illustrative modeling choice, suppose an intersectoral matrix maps one period's state into the next. This autonomous linear model is not, by itself, an empirically established causal description:

$$x_{t+1} = A x_t .$$

Repeated application gives $x_{t+k} = A^k x_t$ for nonnegative integer k. Eigenvectors identify nonzero vectors whose images are scalar multiples of themselves:

$$A v = \lambda v .$$

Here A is square, v is nonzero, and λ is a scalar. The vector v is an eigenvector and λ its eigenvalue. For real eigenpairs the scalar records expansion, contraction, reversal or collapse along that direction.

Worked development: Eigenpairs: directions transformed into multiples of themselves

For a square matrix A, an eigenpair consists of a scalar λ and a vector $v \neq 0$ satisfying

$$A v = \lambda v .$$

For real eigenpairs, a positive eigenvalue preserves orientation, a negative one reverses it, and a zero eigenvalue sends the vector to zero. Thus "keeps its direction" needs qualification. Multiplying an eigenvector by any nonzero scalar gives another eigenvector for the same eigenvalue.

For $A = diag(3,1)$, the coordinate axes are eigendirections. The vector $(1,1)^T$ is **not** an eigenvector: its image $(3,1)^T$ is not a scalar multiple of it.

Numerical acceptance

Floating-point equality needs a tolerance. We reject zero vectors, normalize the candidate vector, and use the scaled residual

$$r = \frac{\| A u - \lambda u \|_2}{\| A \|_2 + |\lambda|}, u = \frac{v}{\| v \|_2} .$$

When the denominator is zero, the residual numerator is used directly. A small residual verifies an approximate eigenpair, **not** that its eigenvalue is dominant. Near repeated eigenvalues, compare eigenspaces rather than signed columns.

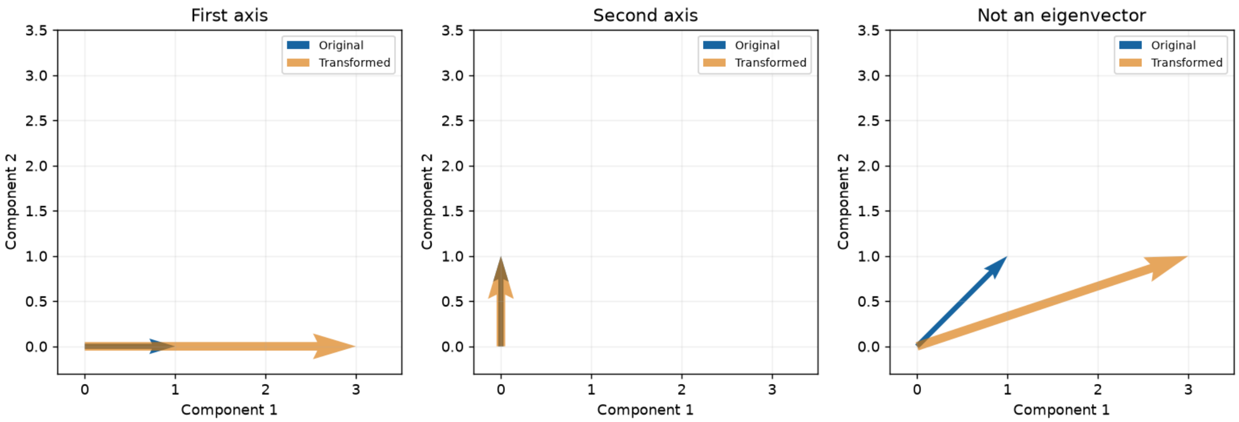

Figure 6.1. Blue original vectors and orange transformed vectors for diag(3, 1). The axes are eigendirections; the diagonal vector is not. Axis overlap is expected.

6.4 Spectral Structure

If a matrix has a complete eigenvector basis,

$$A = P D P^{-1},$$

where (D) contains eigenvalues. Then

$$A^k = P D^k P^{-1}.$$

When a unique eigenvalue has greatest modulus and the initial vector has a suitable component in its mode, that component can dominate relative to the others. Equal dominant moduli, special starts and defective structure require additional care.

For a real symmetric matrix, eigenvalues are real and eigenvectors can be chosen orthonormally. Covariance matrices have this structure, which is why their eigenvectors define principal directions of variation.

The spectral radius is

$$\rho(A) = \max_i |\lambda_i|.$$

For the finite-dimensional autonomous discrete-time system $x_{t+1} = Ax_t$, every initial state tends to zero if and only if $\rho(A) < 1$. At $\rho(A) = 1$, Jordan structure matters. Nonlinear, time-varying or forced systems require additional analysis; continuous-time stability uses a different criterion.

Worked development: Symmetry gives an orthonormal eigenbasis

Take

$$S = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

The characteristic polynomial is $(2-\lambda)^2-1=\lambda^2-4\lambda+3$, so the eigenvalues are 3 and 1. Corresponding unit eigenvectors are $(1,1)^T/\sqrt{2}$ and $(1,-1)^T/\sqrt{2}$. Numerical libraries can reverse either sign.

For a **real symmetric** matrix, the spectral theorem gives $S=QDQ^T$ with $Q^TQ=I$. With repeated eigenvalues, the eigenbasis within a repeated eigenspace is not unique. This theorem does not apply to arbitrary nonsymmetric real matrices.

For $\lambda=3$, solving $(S-3I)v=0$ gives $v_2=v_1$; for $\lambda=1$, $(S-I)v=0$ gives $v_2=-v_1$. Normalizing provides

$$Q=\frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}, D=\begin{pmatrix}3 & 0\\ 0 & 1\end{pmatrix}.$$

The columns are orthonormal, so $Q^{-1}=Q^T$. Verify both $Q^TQ=I$ and $QDQ^T=S$. Sign changes do not alter the eigenspaces. The notebook sorts eigenvalues together with their eigenvector columns; sorting one without the other would destroy the pairing.

Worked development: Diagonalization, powers, and a counterexample

If a square matrix has a complete eigenvector basis over the chosen field, then

$$A=PDP^{-1}, A^k=PD^kP^{-1}.$$

For our symmetric example,

$$S^k=\frac{1}{2}\begin{pmatrix}3^k+1 & 3^k-1\\ 3^k-1 & 3^k+1\end{pmatrix}.$$

This holds also for $k=0$. Determinants are useful for small hand derivations; numerical eigensolvers are preferable to forming a large characteristic polynomial.

Not every matrix is diagonalizable. The Jordan matrix J below has eigenvalue 1 twice but only one independent eigenvector. Its powers contain a growing off-diagonal entry. Near-defective matrices can also yield ill-conditioned eigenvector matrices even when diagonalization is theoretically possible.

At $k=10$, $3^{10}=59049$, so

$$S^{10}=\begin{pmatrix}29525 & 29524\\ 29524 & 29525\end{pmatrix}.$$

The notebook checks the same result through direct powers and $\mathrm{srai\_math}$ spectral powers for $k=0,1,2,10$. The case $k=0$ must yield the identity.

For the defective counterexample,

$$J=\begin{pmatrix}1 & 1\\ 0 & 1\end{pmatrix}, J^k=\begin{pmatrix}1 & k\\ 0 & 1\end{pmatrix}.$$

Writing $J=I+N$ gives $N^2=0$, hence $J^k=I+kN$. The equation $(J-I)v=0$ forces $v_2=0$. Its one-dimensional eigenspace cannot provide a two-vector basis.

A real matrix may need complex eigenpairs: the rotation

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

has eigenvalues i and $-i$. For example, $(1, -i)^T$ is an eigenvector for i. No real line is preserved by this 90-degree rotation.

Worked development: Spectral radius and discrete-time stability

The spectral radius is

$$\rho(A) = \max_i |\lambda_i|.$$

For a finite-dimensional **autonomous discrete-time linear system** $x_{t+1} = A x_t$, every initial state tends to zero exactly when $\rho(A) < 1$. This is not the same criterion as for continuous-time dynamics. Inputs, time-varying coefficients and nonlinearities require other analysis.

At radius one, the identity keeps states constant while a nontrivial Jordan block can cause growth. Even below one, a nonnormal matrix may show substantial transient amplification before eventual decay. A single decaying trajectory cannot establish stability for all initial states.

Compare

$$B = \begin{pmatrix} 0.7 & 0.1 \\ 0 & 0.8 \end{pmatrix}, U = \begin{pmatrix} 1.1 & 0 \\ 0 & 0.9 \end{pmatrix}.$$

The radii are 0.8 and 1.1. For B, starting at $(1, 1)^T$ gives $0.8^t (1, 1)^T$. For U, the special start $(0, 1)^T$ decays despite the unstable first mode. Testing only that start could falsely suggest system-wide stability.

For a diagonalizable stable matrix the powers of each eigenvalue decay. Jordan powers may contain polynomial factors, but exponential decay with modulus below one eventually dominates them. At modulus one, those polynomial factors need not decay. Thus the identity has constant trajectories while J can grow linearly.

The nonnormal matrix

$$T = \begin{pmatrix} 0.8 & 4 \\ 0 & 0.8 \end{pmatrix}$$

has radius 0.8, but the off-diagonal entry of T^k is $4k(0.8)^{k-1}$ for $k \geq 1$. A start at $(0, 1)^T$ initially amplifies before decaying. Long-run stability does not guarantee acceptable short-run exposure.

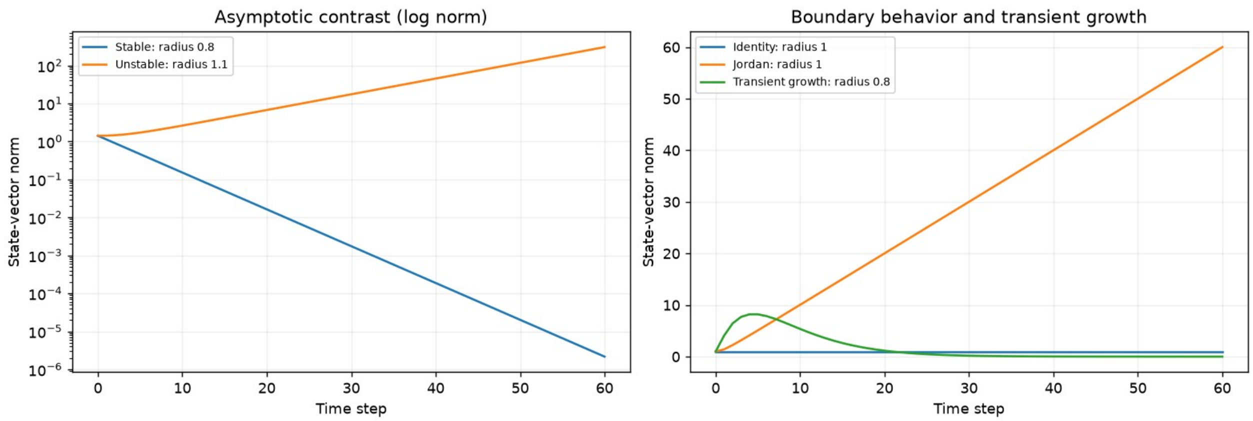

Figure 6.2. Left: stable and unstable trajectories on a log-norm scale. Right: identity, Jordan growth and transient amplification. These are synthetic autonomous linear systems.

6.5 Power Iteration

Starting with a nonzero x_0, provided Ax_k is nonzero at each step, normalized power iteration takes the form:

$$x_{k+1} = \frac{A x_k}{\| A x_k \|},$$

often converges toward a dominant eigenvector. It is useful for large systems, but it normally finds only one mode and can struggle when dominant magnitudes are tied.

Worked development: Power iteration: verify the residual, not just a stable scalar

Repeated multiplication and normalization often select a dominant eigendirection. Standard convergence needs a unique eigenvalue of largest modulus and a suitable initial component, with additional care for nonsymmetric/defective problems. A negative dominant eigenvalue can make vector signs alternate; a residual-based test handles that.

The bundled runtime's legacy power_iteration stops on changes in the Rayleigh quotient and its verify_eigenpair does not reject zero vectors. **This lesson therefore uses the explicit reference functions above for acceptance**, without changing or relabelling the package. These are small dense teaching routines, not production sparse solvers.

For $diag(1, -1)$ and initial vector $(1, 1)$, the Rayleigh quotient remains zero while the vector is not an eigenvector. Conversely, starting on a nondominant eigenvector gives residual zero immediately: residual convergence alone cannot establish dominance.

The explicit reference iteration normalizes every step, checks a scaled residual against 10^{-10} and raises an error if its iteration limit is reached. This is a documented tolerance for these examples, not a universal numerical standard. The routine accepts dense real matrices; the residual checker separately supports complex eigenpairs.

For nonsymmetric matrices, a suitable initial component involves the left/right eigenstructure, not merely an informal Euclidean alignment with one right eigenvector. Spectral separation and conditioning affect convergence. The notebook tests a negative dominant eigenvalue, tied-modulus oscillation and a nondominant initial eigenvector.

The unmodified srai_math 1.1.1rc1 package remains the identifiable prerelease runtime. Its spectral decomposition and matrix-power functions are used directly; the two visible lesson-level safeguards do not redefine the shared package or certify every library function.

6.6 Statistics, AI and Networks

Spectral methods support principal-component analysis, spectral clustering, PageRank, graph diffusion, recurrent-system stability and graph neural networks. Eigenvalues summarize directional strength, but eigenvectors are defined only up to scale and sign; their interpretation must not depend on an arbitrary orientation.

Worked development: Statistics: covariance eigenvectors and principal components

Center each column before forming sample covariance, using divisor $n-1$. For a unit vector q, the sample variance of centered scores $X_c q$ equals $q^T C q$. Covariance eigenvectors identify orthogonal directions of variation, and eigenvalues quantify their variances.

The following four observations illustrate arithmetic only. Scaling a measured variable changes covariance PCA; correlation/standardized PCA answers a different question. High explained variance does not prove prediction accuracy, representativeness, causality or decision usefulness.

With observations in rows, use

$$X=\begin{pmatrix}2 & 1\\ 3 & 2\\ 4 & 2.5\\ 5 & 4\end{pmatrix}, \acute{x}=(3.5, 2.375).$$

Subtract the column means to form X_c, then compute

$$C=\frac{X_c^T X_c}{n-1}=\begin{pmatrix}5/3 & 19/12\\ 19/12 & 25/16\end{pmatrix}.$$

The descending eigenvalues are approximately 3.19877307 and 0.03039360; the variance ratios are 0.99058779 and 0.00941221. Their sum is one, while the eigenvalues sum to $tr(C)=155/48$.

For a unit covariance eigenvector q, the sample variance of $X_c q$ is $q^T C q=\lambda$. The notebook verifies this identity and reconstructs the centered data using both components. With selected orthonormal columns Q_r, reconstruction is $X_c Q_r Q_r^T$: the matrix $Q_r Q_r^T$ is the orthogonal projector studied in Lesson 5.

PCA maximizes retained variance, not prediction accuracy, causal relevance or policy value. Multiplying a variable by 100 changes covariance PCA. Standardization and correlation PCA may be appropriate, but answer a different question and must be stated.

Graph interpretations likewise depend on the chosen operator. Spectral clustering may use a graph Laplacian, while PageRank uses an explicitly stochastic model with teleportation. An arbitrary sector matrix is neither. Local linear diagnostics for nonlinear recurrent models do not automatically establish global stability.

6.7 AI Reasoning Clinic

Statement: “The dominant eigenvector identifies the most important policy.”

Evaluation: It identifies a dominant mathematical mode under the specified matrix. “Importance” requires substantive interpretation, validated matrix construction and a clear decision objective.

Worked development: Decision interpretation: a synthetic sector propagation model

We posit $x_{t+1} = M x_t$, where entry M_{ij} maps component j into component i. These coefficients are invented for teaching; they are not fitted to observations. Rows and columns do **not** sum to one: this is not a Markov transition-probability matrix.

This strictly positive matrix has a positive dominant eigendirection. Normalizing that eigenvector to sum one makes its components easier to compare, but does not turn them into budget allocations or causal importance. Its dominant eigenvalue exceeds one, implying amplification along that mode within this hypothetical autonomous model.

The tested example uses the ordering Agriculture, Health, Energy and Transport:

$$M = \begin{pmatrix} 0.70 & 0.10 & 0.20 & 0.15 \\ 0.10 & 0.80 & 0.25 & 0.10 \\ 0.25 & 0.10 & 0.75 & 0.30 \\ 0.20 & 0.15 & 0.35 & 0.70 \end{pmatrix}.$$

Its row sums are $(1.15, 1.25, 1.40, 1.40)$ and column sums $(1.25, 1.15, 1.55, 1.25)$: neither convention makes it stochastic. The spectral radius is approximately 1.31148205. The corresponding positive eigenvector, normalized to sum one, is

$$w \approx (0.20024812, 0.23284988, 0.28255681, 0.28434520)^T.$$

The positive matrix has a positive dominant mode. Its magnitude grows by about 1.31148 per model time step along that direction. Relative direction does not imply convergence to a finite nonzero steady state. The notebook compares the iterative eigenvalue to a dense eigensolver and checks the residual.

Transport’s and Energy’s larger components do not establish optimal spending or causal leverage. Data validity, time scale, intervention effects, objectives, costs, constraints and uncertainty are not supplied by eigenvector normalization.

A defensible statement is: "Under this specified synthetic model, the reported mode amplifies and its eigenpair residual is small. Operational use would require defensible coefficients, validation, sensitivity analysis and an explicit decision objective."

6.8 SRAI Principle No. 6

A dominant mathematical direction is not automatically a desirable social direction.

Mathematics answers a conditional question: what follows from this matrix and rule? Accountable practice also asks why this representation is appropriate, which outcomes matter, and which actions are feasible and justified. Reproducibility supports that judgment; it does not replace it.

6.9 Exercises and Laboratory

Exercises — attempt these before the solutions

Exercise 1

For $diag(-2,0)$, identify two eigenpairs and interpret them. Why is the zero vector excluded?

Exercise 2

Derive the characteristic polynomial of S, and find an orthonormal eigenbasis.

Exercise 3

Compute S^{10} by its eigenbasis. What happens for power zero?

Exercise 4

Prove $J^k = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$. Why can J not be diagonalized?

Exercise 5

Explain why equal successive Rayleigh quotients do not imply convergence for $diag(1,-1)$.

Exercise 6

Give a starting vector that hides the unstable mode of $diag(1.1,0.9)$.

Exercise 7

Explain why radius 0.8 does not rule out short-term amplification.

Exercise 8

Compute the two sample covariance eigenvalues and explain the denominator $n-1$.

Exercise 9

Interpret the four-sector weights and list three reasons they cannot determine expenditure priorities.

Exercise 10

State the difference between an approximate eigenpair residual check and proof of dominance.

Worked solutions and interpretation

Solution 1

$(\lambda, v)=\left(-2,(1,0)^T\right)$ reverses orientation and doubles length; $\left(0,(0,1)^T\right)$ maps to zero. Allowing $v=0$ would satisfy the equation for every scalar and identify no direction.

Solution 2

$det(S-\lambda I)=(2-\lambda)^2-1=(\lambda-3)(\lambda-1)$. Normalize $(1,1)^T$ and $(1,-1)^T$ by $\sqrt{2}$; they are orthogonal.

Solution 3

The diagonal entries are $(59049+1)/2=29525$ and off-diagonal entries $(59049-1)/2=29524$. Power zero yields the identity.

Solution 4

Write $J=I+N$ with $N^2=0$; the binomial expansion gives $I+kN$. The eigenspace for eigenvalue one has dimension one, not two.

Solution 5

Equal-magnitude components give Rayleigh quotient zero while the sign of one component alternates. The eigenpair residual remains nonzero.

Solution 6

Starting at $(0,1)^T$ produces decay as 0.9^t, despite the unstable eigenvalue 1.1. Stability for all starts is a stronger claim.

Solution 7

For the upper-triangular example, coupling feeds the second component into the first. The off-diagonal term in its kth power is $4k(0.8)^{k-1}$, which can initially increase but eventually vanishes.

Solution 8

The covariance eigenvalues are approximately 3.19877307 and 0.03039360. Sample covariance uses $n-1$ after estimating the mean; this is not a claim that four observations support reliable statistical inference.

Solution 9

The normalized components describe one model eigendirection. Coefficients are synthetic, causal identification is absent, and costs/objectives/constraints are not modeled. All three prevent interpreting weights as allocations.

Solution 10

Any eigenpair can have residual zero, including a nondominant one. Dominance requires spectral comparison or justified convergence assumptions; the sector example also compares against the dense eigensolver.

Laboratory protocol: run the exact candidate from its setup cells through the final PASS. Retain the environment and executed copy separately. Inspect equations, figures and tables. Do not disable checks or alter hashes to obtain a green result. A sign difference is not the same as an incorrect eigenspace or a failed residual.

6.10 Key Insight

Eigenvalues and eigenvectors expose invariant directions, dominant dynamics, variance structure and stability hidden inside matrices.

6.11 Looking Ahead

Carry forward three habits: state assumptions, verify calculations through independent identities, and separate numerical structure from substantive meaning. Spectral reasoning will recur in dimensionality reduction, optimization, graph methods and stability analysis.

Control note and chapter-to-notebook alignment

Chapter 6.3 corresponds to notebook section 1; spectral structure in 6.4 to notebook sections 2, 3 and 5; iteration in 6.5 to notebook section 4; statistics in 6.6 to section 6; the sector clinic in 6.7 to section 7; and exercises in 6.9 to notebook sections 8–9. The same matrices, conventions, answer values and caveats are used.

The candidate notebook previously passed 26 local mathematical/control tests, static markup validation and sequential plain-Python execution. The user subsequently reported error-free runs in VS Code and Colab. Executed files, environment logs and a separate visual-markup confirmation were not supplied with that message; this chapter does not invent them.

Owner chapter approval, qualified mathematical review, runtime acceptance and publication authorization remain separate gates. The historical “9.x” source heading numbers have been corrected to “6.x” without changing the chapter identity. Six original native displayed equations are retained; damaged inline formulas and ambiguous claims are repaired. The five brief source exercises are expanded into the ten notebook-aligned exercises and solutions.

Further study: Gilbert Strang, Introduction to Linear Algebra; David C. Lay, Steven R. Lay and Judi J. McDonald, Linear Algebra and Its Applications; Lloyd N. Trefethen and David Bau III, Numerical Linear Algebra. These are background references, not a claim of external review of this candidate.